

Fullstack Development

API Architectures and Design #4

Content

- What is API?
- API Architecture Styles
- RESTful API design
- API Security
- **API Testing**

API Testing

Functionality <Cypress, Postman, Insomnia>

- Endpoints return the correct data and perform the expected operations

Reliability

- Ability to handle various scenarios (error conditions) without crashing

API Testing (2)

Security <OWASP Tools, Google Apigee>

- Identifying **vulnerabilities**
- Unauthorized access, data breaches, injection flaw

Performance <Apache JMeter, K6>

- Measuring response time and throughput under different **load conditions**

API Performance Testing

Performance Testing Tool

K6

- An open-source **load testing tool** developed by **Grafana Labs**
- Write tests in `JavaScript` or `TypeScript`
 - Run locally (`Windows` , `macOS` , `Linux` , `Docker`)
 - Run on **Grafana Cloud**
- Supports `RESTful` , `GraphQL` , `WebSocket` , `gRPC`
- Supports different types of testing

Types of Testing



K6 Results



What does performance mean?

- Influences the **type of tests** you should perform
- Define normal API traffic and **acceptable response time**



Specify Thresholds



Stress Testing



Spike Testing



Soak Testing



References

- [API Testing: A Guide for Beginners and Expert](#)
- [Is Your API actually ready for user traffic?](#)
- [Git - Template to use TypeScript with k6](#)
- [Getting Started with Performance Testing in Typescript Using K6](#)